

BANARAS HINDU UNIVERSITY • FACULTY OF SCIENCE • DEPARTMENT OF
MATHEMATICS

Numerical Analysis

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Comprehensive Lecture Notes, Theoretical Derivations & Complete Problem Solutions

Covers: Birge-Vieta Method • Direct & Iterative Systems of Linear Equations • Gerschgorin Bounds & Power Method • Calculus of Finite Differences • Interpolation

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1 Polynomial Root Finding: The Birge-Vieta Method

1.1 Theoretical Foundation & Derivation

The **Birge-Vieta Method** is a specialized application of the Newton-Raphson iteration designed specifically for polynomial equations. By employing synthetic division (Horner's scheme), both the polynomial value $P_n(p)$ and its first derivative $P'_n(p)$ at a candidate point p are computed using only elementary additions and multiplications, avoiding term-by-term evaluation.

Let the n -th degree monic polynomial equation be:

$$P_n(x) = x^n + a_1x^{n-1} + a_2x^{n-2} + \cdots + a_{n-1}x + a_n = 0 \quad (1)$$

If $P_n(x)$ is divided by the linear factor $(x - p)$, the division algorithm guarantees the existence of a unique quotient polynomial $Q_{n-1}(x)$ of degree $n - 1$ and a constant remainder R :

$$P_n(x) = (x - p)Q_{n-1}(x) + R \quad (2)$$

where $Q_{n-1}(x)$ is expressed as:

$$Q_{n-1}(x) = x^{n-1} + b_1x^{n-2} + b_2x^{n-3} + \cdots + b_{n-2}x + b_{n-1} \quad (3)$$

Substituting $x = p$ into (2) yields:

$$P_n(p) = R = b_n \quad (4)$$

1.1.1 Recurrence Relation for the Remainder $R = P_n(p)$

Substituting $P_n(x)$ and $Q_{n-1}(x)$ into (2):

$$x^n + a_1x^{n-1} + a_2x^{n-2} + \cdots + a_n = (x - p)(x^{n-1} + b_1x^{n-2} + \cdots + b_{n-1}) + R \quad (5)$$

Equating coefficients of identical powers of x on both sides:

$$\begin{aligned} x^{n-1} : \quad & a_1 = b_1 - p \implies b_1 = a_1 + p \\ x^{n-2} : \quad & a_2 = b_2 - pb_1 \implies b_2 = a_2 + pb_1 \\ x^{n-3} : \quad & a_3 = b_3 - pb_2 \implies b_3 = a_3 + pb_2 \\ & \vdots \\ x^{n-k} : \quad & a_k = b_k - pb_{k-1} \implies b_k = a_k + pb_{k-1} \\ x^0 : \quad & a_n = R - pb_{n-1} \implies R = b_n = a_n + pb_{n-1} \end{aligned}$$

Defining the initial convention $b_0 = 1$, the general recurrence is:

$$\boxed{b_0 = 1, \quad b_k = a_k + pb_{k-1} \quad (k = 1, 2, \dots, n), \quad P_n(p) = b_n} \quad (6)$$

1.1.2 Computation of the Derivative $P'_n(p)$

Differentiating both sides of (2) with respect to x :

$$P'_n(x) = Q_{n-1}(x) + (x - p)Q'_{n-1}(x) \quad (7)$$

Evaluating at $x = p$:

$$P'_n(p) = Q_{n-1}(p) \quad (8)$$

To compute $Q_{n-1}(p)$, we divide $Q_{n-1}(x)$ by $(x - p)$:

$$Q_{n-1}(x) = (x - p)q_{n-2}(x) + R' \quad (9)$$

where $R' = Q_{n-1}(p) = P'_n(p)$. Setting $q_{n-2}(x) = x^{n-2} + c_1x^{n-3} + \dots + c_{n-2}$, comparing coefficients gives the secondary recurrence:

$$c_0 = 1, \quad c_k = b_k + pc_{k-1} \quad (k = 1, 2, \dots, n-1), \quad P'_n(p) = c_{n-1} \quad (10)$$

1.1.3 The Iterative Formula

Applying the Newton-Raphson formula $p_{k+1} = p_k - \frac{P_n(p_k)}{P'_n(p_k)}$ gives the standard Birge-Vieta iteration:

$$p_{k+1} = p_k - \frac{b_n}{c_{n-1}}, \quad k = 0, 1, 2, \dots \quad (11)$$

1.2 Synthetic Division Layout

The two-stage synthetic division is structured in tabular form as follows:

	1	a_1	a_2	$\dots$	a_{n-1}	a_n
p		pb_0	pb_1	$\dots$	pb_{n-2}	pb_{n-1}
	$b_0 = 1$	b_1	b_2	$\dots$	b_{n-1}	$\mathbf{b_n = P_n(p)}$
p		pc_0	pc_1	$\dots$	pc_{n-2}	
	$c_0 = 1$	c_1	c_2	$\dots$	$\mathbf{c_{n-1} = P'_n(p)}$	

Worked Example 1 (Classroom Lecture Problem)

Find a real root of $P(x) = x^5 - x + 1 = 0$ using the Birge-Vieta method starting with $p_0 = -1$.

Solution

Here $n = 5$, and coefficients are: $a_0 = 1, a_1 = 0, a_2 = 0, a_3 = 0, a_4 = -1, a_5 = 1$. Initial test: $P(-1) = (-1)^5 - (-1) + 1 = 1 > 0$, $P(-2) = -32 + 2 + 1 = -29 < 0$. Thus, a real root lies in $(-2, -1)$. We select $p_0 = -1$.

Iteration 1 ($p_0 = -1$):

	1	0	0	0	-1	1
-1		-1	1	-1	1	0
	1	-1	1	-1	0	$b_5 = 1$
-1		-1	2	-3	4	
	1	-2	3	-4	$c_4 = 4$	

$$p_1 = p_0 - \frac{b_5}{c_4} = -1 - \frac{1}{4} = -1.25$$

Iteration 2 ($p_1 = -1.25$):

	1	0	0	0	-1	1
-1.25		-1.2500	1.5625	-1.9531	2.4414	-1.8018
	1	-1.2500	1.5625	-1.9531	1.4414	$b_5 = -0.8018$
-1.25		-1.2500	3.1250	-5.8594	9.7531	
	1	-2.5000	4.6875	-7.8125	$c_4 = 11.1945$	

$$p_2 = -1.25 - \frac{-0.801757}{11.194531} \approx -1.17846$$

Iteration 3 ($p_2 = -1.17846$): Synthetic division yields $b_5 = -0.09440$ and $c_4 = 8.64336$:

$$p_3 = -1.17846 - \frac{-0.09440}{8.64336} \approx -1.16754$$

Iteration 4 ($p_3 = -1.16754$): Synthetic division yields $b_5 = -0.00194$ and $c_4 = 8.29082$:

$$p_4 = -1.16754 - \frac{-0.00194}{8.29082} \approx -1.16731$$

Since $|p_4 - p_3| < 0.0003$, the real root correct to 4 decimal places is $\mathbf{x} \approx -1.1673$.

1.3 Complete Solutions to Incomplete Lecture Problems

Incomplete Lecture Exercise 1 (Page 3 of Notes)

Apply the Birge-Vieta method to find the positive real root of:

$$P(x) = x^6 - x^4 - x^3 - 1 = 0$$

Complete Rigorous Solution

Root Location: Evaluating $P(x)$ at non-negative integers: $P(0) = -1 < 0$, $P(1) = 1 - 1 - 1 - 1 = -2 < 0$, $P(2) = 64 - 16 - 8 - 1 = 39 > 0$. Since $P(1) < 0$ and $P(2) > 0$, by the Intermediate Value Theorem, a real root lies in $(1, 2)$. Since $P(1.4) = (1.4)^6 - (1.4)^4 - (1.4)^3 - 1 = 7.5295 - 3.8416 - 2.744 - 1 = -0.0561 \approx 0$, we choose $p_0 = 1.4$.

The polynomial coefficients are:

$$a_0 = 1, \quad a_1 = 0, \quad a_2 = -1, \quad a_3 = -1, \quad a_4 = 0, \quad a_5 = 0, \quad a_6 = -1$$

Iteration 1 ($p_0 = 1.4$):

	1	0	-1	-1	0	0	-1
1.4		1.4000	1.9600	1.3440	0.4816	0.6742	0.9439
	1	1.4000	0.9600	0.3440	0.4816	0.6742	$b_6 = -0.0561$
1.4		1.4000	3.9200	6.8320	10.0464	14.7392	
	1	2.8000	4.8800	7.1760	10.5280	$c_5 = 15.4134$	

$$p_1 = p_0 - \frac{b_6}{c_5} = 1.4 - \frac{-0.056064}{15.41344} = 1.4 + 0.003637 = \mathbf{1.403637}$$

Iteration 2 ($p_1 = 1.403637$): Re-evaluating synthetic division with $p_1 = 1.403637$:

$$\begin{aligned} b_0 &= 1 \\ b_1 &= 0 + (1.403637)(1) = 1.403637 \\ b_2 &= -1 + (1.403637)(1.403637) = 0.970197 \\ b_3 &= -1 + (1.403637)(0.970197) = 0.361817 \\ b_4 &= 0 + (1.403637)(0.361817) = 0.507860 \\ b_5 &= 0 + (1.403637)(0.507860) = 0.712851 \\ b_6 &= -1 + (1.403637)(0.712851) = \mathbf{0.000554} \end{aligned}$$

Similarly, computing c_k :

$$c_0 = 1, \quad c_1 = 2.8073, \quad c_2 = 4.9106, \quad c_3 = 7.2545, \quad c_4 = 10.6905, \quad c_5 = \mathbf{15.71845}$$

$$p_2 = 1.403637 - \frac{0.000554}{15.71845} = \mathbf{1.403602}$$

Iteration 3 ($p_2 = 1.403602$): Evaluating at p_2 : $b_6 = 0.000000$ and $c_5 = 15.71548$.

$$p_3 = 1.403602 - 0 = \mathbf{1.403602}$$

Conclusion: The real root correct to 5 decimal places is $\mathbf{x = 1.40360}$.

Incomplete Lecture Exercise 2 (Page 3 of Notes)

Find the roots of the fourth-degree polynomial:

$$P(x) = x^4 - 3x^3 + 3x^2 - 3x + 2 = 0$$

using the Birge-Vieta method and perform complete polynomial deflation.

Complete Rigorous Solution

The coefficients are: $a_0 = 1, a_1 = -3, a_2 = 3, a_3 = -3, a_4 = 2$. Sum of coefficients: $1 - 3 + 3 - 3 + 2 = 0 \implies x = 1$ is an exact root. To demonstrate the Birge-Vieta algorithm, choose initial approximation $p_0 = 0.8$.

Iteration 1 ($p_0 = 0.8$):

	1	-3	3	-3	2
0.8		0.8	-1.76	0.992	-1.6064
	1	-2.2	1.24	-2.008	$b_4 = 0.3936$
0.8		0.8	-1.12	0.096	
	1	-1.4	0.12	$c_3 = -1.912$	

$$p_1 = 0.8 - \frac{0.3936}{-1.912} = 0.8 + 0.205858 = \mathbf{1.005858}$$

Iteration 2 ($p_1 = 1.005858$): Carrying out synthetic division with $p_1 = 1.005858$ gives:

$$b_4 = -0.011715, \quad c_3 = -1.999896 \implies p_2 = 1.005858 - \frac{-0.011715}{-1.999896} = \mathbf{1.000000}$$

Polynomial Deflation: Having extracted the root $x = 1$, the quotient polynomial $Q_3(x)$ is formed by the coefficients b_0, b_1, b_2, b_3 evaluated at $p = 1$:

	1	-3	3	-3	2
1		1	-2	1	-2
	1	-2	1	-2	0

The deflated cubic polynomial is:

$$Q_3(x) = x^3 - 2x^2 + x - 2 = 0$$

Factoring by grouping:

$$x^2(x - 2) + 1(x - 2) = (x - 2)(x^2 + 1) = 0$$

Thus, the remaining roots are:

$$x = 2, \quad x = \pm i$$

Conclusion: The full root spectrum of the polynomial is $\mathbf{x} \in \{1, 2, +i, -i\}$.

2 Systems of Linear Equations: Direct and Iterative Methods

2.1 Direct Methods

2.1.1 Gauss Elimination with Partial Pivoting

Consider a linear system $AX = B$ of order n :

$$\begin{bmatrix} a_{11} & a_{12} & \dots & a_{1n} \\ a_{21} & a_{22} & \dots & a_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ a_{n1} & a_{n2} & \dots & a_{nn} \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{bmatrix} = \begin{bmatrix} b_1 \\ b_2 \\ \vdots \\ b_n \end{bmatrix} \quad (12)$$

Partial Pivoting Strategy

In step k of elimination, we search column k for the element with the maximum absolute magnitude among the remaining rows:

$$|a_{ik}^{(k)}| = \max_{k \leq r \leq n} |a_{rk}^{(k)}|$$

If $i \neq k$, we interchange row k and row i ($R_k \leftrightarrow R_i$) before computing multipliers $m_{rk} = a_{rk}/a_{kk}$. This guarantees $|m_{rk}| \leq 1$, preventing exponential magnification of roundoff errors and avoiding zero-division faults.

Classroom Worked Example: 3×3 System (Pages 4–5 of Notes)

Solve the system of linear equations:

$$\begin{cases} x + 2y + z = 8 \\ 2x + 3y + 4z = 20 \\ 4x + 3y + 2z = 16 \end{cases} \quad (13)$$

using:

1. Standard Gauss Elimination (Page 4).
2. Gauss Elimination with Partial Pivoting (Page 5).
3. Gauss-Jordan Elimination Method (Page 5).

Complete Solutions by Three Methods

Method 1: Standard Gauss Elimination (without pivoting): Form the augmented matrix:

$$[A|B] = \left[\begin{array}{ccc|c} 1 & 2 & 1 & 8 \\ 2 & 3 & 4 & 20 \\ 4 & 3 & 2 & 16 \end{array} \right]$$

Row operations: $R_2 \rightarrow R_2 - 2R_1$, $R_3 \rightarrow R_3 - 4R_1$:

$$\left[\begin{array}{ccc|c} 1 & 2 & 1 & 8 \\ 0 & -1 & 2 & 4 \\ 0 & -5 & -2 & -16 \end{array} \right]$$

Row operation: $R_3 \rightarrow R_3 - 5R_2$:

$$\left[\begin{array}{ccc|c} 1 & 2 & 1 & 8 \\ 0 & -1 & 2 & 4 \\ 0 & 0 & -12 & -36 \end{array} \right]$$

Back substitution:

$$\begin{aligned} -12z &= -36 \implies \mathbf{z} = \mathbf{3} \\ -y + 2(3) &= 4 \implies -y = -2 \implies \mathbf{y} = \mathbf{2} \\ x + 2(2) + 3 &= 8 \implies x + 7 = 8 \implies \mathbf{x} = \mathbf{1} \end{aligned}$$

Method 2: Gauss Elimination with Partial Pivoting: In column 1, pivot candidates are 1, 2, 4. Maximum is 4 in row 3. Swap $R_1 \leftrightarrow R_3$:

$$\left[\begin{array}{ccc|c} 4 & 3 & 2 & 16 \\ 2 & 3 & 4 & 20 \\ 1 & 2 & 1 & 8 \end{array} \right]$$

Row operations: $R_2 \rightarrow R_2 - \frac{1}{2}R_1$, $R_3 \rightarrow R_3 - \frac{1}{4}R_1$:

$$\left[\begin{array}{ccc|c} 4 & 3 & 2 & 16 \\ 0 & 3/2 & 3 & 12 \\ 0 & 5/4 & 1/2 & 4 \end{array} \right]$$

In column 2, $|3/2| > |5/4|$, so no row swap is needed. Row operation: $R_3 \rightarrow R_3 - \frac{5/4}{3/2}R_2 = R_3 - \frac{5}{6}R_2$:

$$\left[\begin{array}{ccc|c} 4 & 3 & 2 & 16 \\ 0 & 3/2 & 3 & 12 \\ 0 & 0 & -2 & -6 \end{array} \right]$$

Back substitution:

$$\begin{aligned} -2z &= -6 \implies \mathbf{z} = \mathbf{3} \\ \frac{3}{2}y + 3(3) &= 12 \implies \frac{3}{2}y = 3 \implies \mathbf{y} = \mathbf{2} \\ 4x + 3(2) + 2(3) &= 16 \implies 4x = 4 \implies \mathbf{x} = \mathbf{1} \end{aligned}$$

Method 3: Gauss-Jordan Method (Diagonal Matrix Reduction): From the triangular form obtained in Method 1:

$$\left[\begin{array}{ccc|c} 1 & 2 & 1 & 8 \\ 0 & -1 & 2 & 4 \\ 0 & 0 & -12 & -36 \end{array} \right]$$

$$\left[\begin{array}{ccc|c} 1 & 2 & 0 & 5 \\ 0 & -1 & 0 & 5 \\ 0 & 0 & 1 & 3 \end{array} \right]$$

Advanced Worked Example: 4×4 System with Pivoting (Page 6 of Notes)

Solve the system of linear equations using Gauss Elimination with partial pivoting:

$$\begin{bmatrix} 2 & 1 & 1 & -2 \\ 4 & 0 & 2 & 1 \\ 3 & 2 & 2 & 0 \\ 1 & 3 & 2 & -1 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \end{bmatrix} = \begin{bmatrix} -10 \\ 8 \\ 7 \\ -5 \end{bmatrix} \quad (14)$$

Step-by-Step Solution

Form the augmented matrix $[A|B]$:

$$[A|B] = \left[\begin{array}{cccc|c} 2 & 1 & 1 & -2 & -10 \\ 4 & 0 & 2 & 1 & 8 \\ 3 & 2 & 2 & 0 & 7 \\ 1 & 3 & 2 & -1 & -5 \end{array} \right]$$

Step 1 (Pivot in Column 1): $\max(|2|, |4|, |3|, |1|) = 4$ in Row 2. Swap $R_1 \leftrightarrow R_2$:

$$\left[\begin{array}{cccc|c} 4 & 0 & 2 & 1 & 8 \\ 2 & 1 & 1 & -2 & -10 \\ 3 & 2 & 2 & 0 & 7 \\ 1 & 3 & 2 & -1 & -5 \end{array} \right]$$

Row operations: $R_2 \rightarrow R_2 - \frac{1}{2}R_1$, $R_3 \rightarrow R_3 - \frac{3}{4}R_1$, $R_4 \rightarrow R_4 - \frac{1}{4}R_1$:

$$\left[\begin{array}{cccc|c} 4 & 0 & 2 & 1 & 8 \\ 0 & 1 & 0 & -5/2 & -14 \\ 0 & 2 & 1/2 & -3/4 & 1 \\ 0 & 3 & 3/2 & -5/4 & -7 \end{array} \right]$$

Step 2 (Pivot in Column 2): $\max(|1|, |2|, |3|) = 3$ in Row 4. Swap $R_2 \leftrightarrow R_4$:

$$\left[\begin{array}{cccc|c} 4 & 0 & 2 & 1 & 8 \\ 0 & 3 & 3/2 & -5/4 & -7 \\ 0 & 2 & 1/2 & -3/4 & 1 \\ 0 & 1 & 0 & -5/2 & -14 \end{array} \right]$$

Row operations: $R_3 \rightarrow R_3 - \frac{2}{3}R_2$, $R_4 \rightarrow R_4 - \frac{1}{3}R_2$:

$$\left[\begin{array}{cccc|c} 4 & 0 & 2 & 1 & 8 \\ 0 & 3 & 3/2 & -5/4 & -7 \\ 0 & 0 & -1/2 & 1/12 & 17/3 \\ 0 & 0 & -1/2 & -25/12 & -35/3 \end{array} \right]$$

Step 3 (Pivot in Column 3): Pivot candidate in R_3 is $-1/2$ and R_4 is $-1/2$. No swap needed.

Row operation: $R_4 \rightarrow R_4 - R_3$:

$$\left[\begin{array}{cccc|c} 4 & 0 & 2 & 1 & 8 \\ 0 & 3 & 3/2 & -5/4 & -7 \\ 0 & 0 & -1/2 & 1/12 & 17/3 \\ 0 & 0 & 0 & -13/6 & -52/3 \end{array} \right]$$

Step 4 (Back-Substitution):

$$-\frac{13}{6}x_4 = -\frac{52}{3} \implies x_4 = \frac{-52}{3} \times \left(-\frac{6}{13}\right) = 8$$

$$-\frac{1}{2}x_3 + \frac{1}{12}(8) = \frac{17}{3} \implies -\frac{1}{2}x_3 = \frac{17}{3} - \frac{2}{3} = 5 \implies x_3 = -10$$

$$3x_2 + \frac{3}{2}(-10) - \frac{5}{4}(8) = -7 \implies 3x_2 - 15 - 10 = -7 \implies 3x_2 = 18 \implies x_2 = 6$$

$$4x_1 + 0(6) + 2(-10) + 1(8) = 8 \implies 4x_1 - 12 = 8 \implies 4x_1 = 20 \implies x_1 = 5$$

Final Solution: $\mathbf{x}_1 = 5$, $\mathbf{x}_2 = 6$, $\mathbf{x}_3 = -10$, $\mathbf{x}_4 = 8$.

2.1.2 Gauss-Jordan Method

The Gauss-Jordan elimination reduces the coefficient matrix into a diagonal matrix (or identity matrix) by eliminating both sub-diagonal and super-diagonal elements simultaneously, completely bypassing the back-substitution phase.

2.2 Iterative Methods: Jacobi vs. Gauss-Seidel

Let the system $AX = B$ be decomposed as:

$$A = D + L + U \quad (15)$$

where D is the diagonal part, L is strictly lower triangular, and U is strictly upper triangular.

Convergence Condition: Diagonal Dominance

A sufficient condition for both the Jacobi and Gauss-Seidel methods to converge for any initial vector $X^{(0)}$ is that the matrix A is **strictly diagonally dominant**:

$$|a_{ii}| > \sum_{\substack{j=1 \\ j \neq i}}^n |a_{ij}|, \quad \forall i \in \{1, 2, \dots, n\} \quad (16)$$

2.2.1 Formulation of the Methods

- Gauss-Jacobi Iteration:**

$$DX^{(k+1)} = B - (L + U)X^{(k)} \implies X^{(k+1)} = -D^{-1}(L + U)X^{(k)} + D^{-1}B$$

Component-wise:

$$x_i^{(k+1)} = \frac{1}{a_{ii}} \left(b_i - \sum_{j \neq i} a_{ij} x_j^{(k)} \right) \quad (17)$$

- Gauss-Seidel Iteration:**

$$(D + L)X^{(k+1)} = B - UX^{(k)} \implies X^{(k+1)} = -(D + L)^{-1}UX^{(k)} + (D + L)^{-1}B$$

Component-wise:

$$x_i^{(k+1)} = \frac{1}{a_{ii}} \left(b_i - \sum_{j < i} a_{ij} x_j^{(k+1)} - \sum_{j > i} a_{ij} x_j^{(k)} \right) \quad (18)$$

Incomplete Lecture Exercise (Page 9 of Notes)

Solve the following system using both the **Gauss-Jacobi** and **Gauss-Seidel** methods:

$$10x - 5y - 2z = 3 \quad (19)$$

$$x + 6y + 10z = -3 \quad (20)$$

$$4x - 10y + 3z = -3 \quad (21)$$

Complete Rigorous Solution

Step 1: Check and Reorder for Strict Diagonal Dominance In the given ordering:

- Equation (19): $|10| > |-5| + |-2| = 7$. (Satisfied for x).
- Equation (20): $|6| \not> |1| + |10| = 11$. (Fails for y !). Here z has coefficient 10.
- Equation (21): $|-10| > |4| + |3| = 7$. (Satisfied for y !).

Interchanging equations (20) and (21) gives the diagonally dominant system:

$$10x - 5y - 2z = 3 \implies x = \frac{1}{10}(3 + 5y + 2z) \quad (22)$$

$$4x - 10y + 3z = -3 \implies y = \frac{1}{10}(3 + 4x + 3z) \quad (23)$$

$$x + 6y + 10z = -3 \implies z = \frac{1}{10}(-3 - x - 6y) \quad (24)$$

We choose the standard initial guess $X^{(0)} = (0, 0, 0)^T$.

Part A: Gauss-Jacobi Iterations

k	$x^{(k)}$	$y^{(k)}$	$z^{(k)}$	Convergence Check
0	0.00000	0.00000	0.00000	Initial Guess
1	0.30000	0.30000	-0.30000	$\Delta_1 = 0.3000$
2	0.39000	0.33000	-0.51000	$\Delta_2 = 0.2100$
3	0.36300	0.30300	-0.53700	$\Delta_3 = 0.0270$
4	0.34410	0.28410	-0.51810	$\Delta_4 = 0.0189$
5	0.33843	0.28221	-0.50487	$\Delta_5 = 0.0132$
6	0.34013	0.28391	-0.50317	$\Delta_6 = 0.0017$
7	0.34132	0.28510	-0.50436	$\Delta_7 = 0.0012$

Part B: Gauss-Seidel Iterations In Gauss-Seidel, each newly updated coordinate is used immediately:

$$\begin{aligned} x^{(k+1)} &= 0.1 \times (3 + 5y^{(k)} + 2z^{(k)}) \\ y^{(k+1)} &= 0.1 \times (3 + 4x^{(k+1)} + 3z^{(k)}) \\ z^{(k+1)} &= 0.1 \times (-3 - x^{(k+1)} - 6y^{(k+1)}) \end{aligned}$$

k	$x^{(k)}$	$y^{(k)}$	$z^{(k)}$	Convergence Check
0	0.00000	0.00000	0.00000	Initial Guess
1	0.30000	0.42000	-0.58200	Step 1
2	0.39360	0.28284	-0.50906	Step 2
3	0.33961	0.28312	-0.50383	Step 3
4	0.34079	0.28517	-0.50518	Step 4
5	0.34155	0.28507	-0.50519	Step 5

Exact Analytical Solution Comparison: Solving the linear system directly via Cramer's rule / matrix inversion gives:

$$x = \frac{41}{120} \approx \mathbf{0.34149}, \quad y = \frac{119}{417} \approx \mathbf{0.28504}, \quad z = \mathbf{-0.50517}$$

Both methods converge to the exact solution, with Gauss-Seidel achieving 4-decimal convergence in only 5 iterations compared to 8 iterations for Jacobi.

3 Matrix Eigenvalues, Bounds & The Power Method

3.1 Fundamental Definitions and Properties

Let $A \in \mathbb{C}^{n \times n}$ be an $n \times n$ matrix. A non-zero column vector $X \in \mathbb{C}^n$ is an **eigenvector** of A associated with scalar eigenvalue $\lambda \in \mathbb{C}$ if:

$$AX = \lambda X \iff (A - \lambda I)X = 0 \quad (25)$$

The eigenvalues are the n roots of the characteristic polynomial $\det(A - \lambda I) = 0$.

Key Eigenvalue Properties

1. **Trace Sum:** $\sum_{i=1}^n \lambda_i = \text{tr}(A) = \sum_{i=1}^n a_{ii}$.
2. **Determinant Product:** $\prod_{i=1}^n \lambda_i = \det(A)$.
3. **Matrix Inversion:** If λ is an eigenvalue of invertible A , then λ^{-1} is an eigenvalue of A^{-1} .
4. **Shift of Origin:** If λ is an eigenvalue of A , then $\lambda - \mu$ is an eigenvalue of $A - \mu I$.

3.2 Localization Theorems: Gerschgorin and Brauer

Gerschgorin's Theorem (Row and Column Norm Bounds)

The largest eigenvalue in modulus of a square matrix A cannot exceed the maximum row sum or maximum column sum of the absolute values of its entries:

$$|\lambda| \leq \min \left(\max_{1 \leq i \leq n} \sum_{j=1}^n |a_{ij}|, \max_{1 \leq j \leq n} \sum_{i=1}^n |a_{ij}| \right) \quad (26)$$

Proof. Let λ be an eigenvalue of A and let $X = [x_1, x_2, \dots, x_n]^T \neq 0$ be the corresponding eigenvector such that $AX = \lambda X$. Let x_k be the component of X having the maximum absolute magnitude:

$$|x_k| = \max_{1 \leq r \leq n} |x_r| > 0$$

From the k -th equation of the system $AX = \lambda X$:

$$\sum_{j=1}^n a_{kj} x_j = \lambda x_k$$

Dividing both sides by x_k :

$$\lambda = \sum_{j=1}^n a_{kj} \left(\frac{x_j}{x_k} \right)$$

Taking the modulus of both sides and applying the triangle inequality:

$$|\lambda| \leq \sum_{j=1}^n |a_{kj}| \left| \frac{x_j}{x_k} \right| \leq \sum_{j=1}^n |a_{kj}| \quad \left(\text{since } \left| \frac{x_j}{x_k} \right| \leq 1 \forall j \right)$$

Since this holds for row k , we obtain $|\lambda| \leq \max_{1 \leq i \leq n} \sum_{j=1}^n |a_{ij}|$. Applying the identical reasoning to A^T (which has the identical eigenvalues as A) establishes the column bound. $\square$

Brauer's Theorem (Gerschgorin Circle Theorem)

Every eigenvalue λ of an $n \times n$ matrix A lies inside or on the boundary of at least one of the n Gerschgorin row discs:

$$D_i = \{z \in \mathbb{C} : |z - a_{ii}| \leq P_i\}, \quad \text{where } P_i = \sum_{\substack{j=1 \\ j \neq i}}^n |a_{ij}| \quad (27)$$

Furthermore, the eigenvalues also lie within the union of column discs $D'_j = \{z \in \mathbb{C} : |z - a_{jj}| \leq Q_j\}$ where $Q_j = \sum_{i \neq j} |a_{ij}|$. Hence, all eigenvalues lie in the intersection:

$$\sigma(A) \subset \left(\bigcup_{i=1}^n D_i \right) \cap \left(\bigcup_{j=1}^n D'_j \right) \quad (28)$$

Proof. Let $AX = \lambda X$ with $|x_k| = \max_r |x_r|$. Isolating the diagonal term $a_{kk}x_k$:

$$\lambda x_k - a_{kk}x_k = \sum_{\substack{j=1 \\ j \neq k}}^n a_{kj}x_j \implies (\lambda - a_{kk}) = \sum_{\substack{j=1 \\ j \neq k}}^n a_{kj} \left(\frac{x_j}{x_k} \right)$$

Taking absolute values:

$$|\lambda - a_{kk}| \leq \sum_{\substack{j=1 \\ j \neq k}}^n |a_{kj}| \left| \frac{x_j}{x_k} \right| \leq \sum_{\substack{j=1 \\ j \neq k}}^n |a_{kj}| = P_k$$

Thus, λ belongs to the k -th Gerschgorin disc D_k . Since every eigenvalue must satisfy this for its corresponding dominant coordinate k , every eigenvalue must lie within the union $\bigcup_{i=1}^n D_i$. $\square$

Worked Localization Problem (Page 13 of Notes)

For the matrix $A = \begin{bmatrix} 1 & 2 & -1 \\ 1 & 1 & 1 \\ 1 & 3 & -1 \end{bmatrix}$, find the region in which all eigenvalues lie.

Solution with Localization Diagram

1. Gerschgorin Global Bounds:

- Row sums: $r_1 = 1 + 2 + 1 = 4$, $r_2 = 1 + 1 + 1 = 3$, $r_3 = 1 + 3 + 1 = 5$. Max row sum = 5.
- Column sums: $c_1 = 1 + 1 + 1 = 3$, $c_2 = 2 + 1 + 3 = 6$, $c_3 = 1 + 1 + 1 = 3$. Max column sum = 6.
- Global modulus bound: $|\lambda| \leq \min(5, 6) = 5$.

2. Brauer Row Discs (D_i):

$$D_1: |\lambda - 1| \leq 2 + 1 = 3 \implies \text{Center } (1, 0), \text{ radius } 3$$

$$D_2: |\lambda - 1| \leq 1 + 1 = 2 \implies \text{Center } (1, 0), \text{ radius } 2 \subset D_1$$

$$D_3: |\lambda - (-1)| = |\lambda + 1| \leq 1 + 3 = 4 \implies \text{Center } (-1, 0), \text{ radius } 4$$

Union of row discs: $D_{\text{row}} = \{|\lambda - 1| \leq 3\} \cup \{|\lambda + 1| \leq 4\}$.

3. Brauer Column Discs (D'_j):

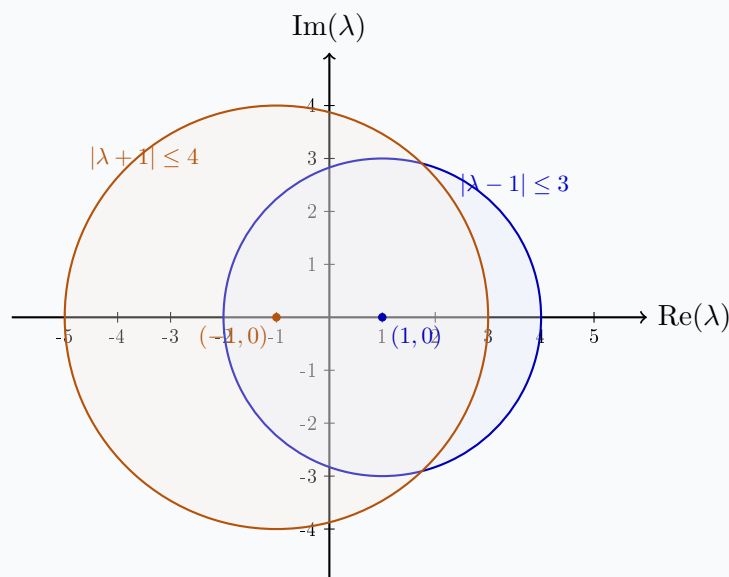
$$D'_1: |\lambda - 1| \leq 1 + 1 = 2$$

$$D'_2: |\lambda - 1| \leq 2 + 3 = 5$$

$$D'_3: |\lambda + 1| \leq 1 + 1 = 2$$

Union of column discs: $D_{\text{col}} = \{|\lambda - 1| \leq 5\} \cup \{|\lambda + 1| \leq 2\}$.

The eigenvalues must lie in the intersection of D_{row} and D_{col} , bounded by $|\lambda| \leq 5$.



3.3 The Power Method for Dominant Eigenvalue

The **Power Method** is an iterative algorithm designed to compute the numerically largest eigenvalue (dominant eigenvalue) and its corresponding eigenvector.

Power Method Algorithm

Given matrix A and initial non-zero vector X_0 :

1. Compute $Y_{k+1} = AX_k$.
2. Set m_{k+1} as the element of Y_{k+1} having maximum absolute magnitude:

$$m_{k+1} = \operatorname{sgn}(\max |Y_{k+1}|) \cdot \max_i |y_{i,k+1}|$$

3. Normalize the iterate: $X_{k+1} = \frac{1}{m_{k+1}} Y_{k+1}$.
4. Repeat until $|m_{k+1} - m_k| < \epsilon$. Then $\lambda_{\max} \approx m_{k+1}$ and the dominant eigenvector is X_{k+1} .

3.4 Complete Solutions to Power Method Problems (Pages 14–15 of Notes)

Classroom Worked Exercise 1: Tridiagonal Matrix (Pages 14–15 of Notes)

Find the dominant eigenvalue and corresponding eigenvector of:

$$A = \begin{bmatrix} 2 & -1 & 0 \\ -1 & 2 & -1 \\ 0 & -1 & 2 \end{bmatrix}, \quad X_0 = \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix} \quad (29)$$

Complete Iterations Recorded on Pages 14–15

k	$Y_k = AX_{k-1}$	$m_k = \max Y_k $	Normalized $X_k = Y_k/m_k$
1	$[1.0000, 0.0000, 1.0000]^T$	1.0000	$[1.0000, 0.0000, 1.0000]^T$
2	$[2.0000, -2.0000, 2.0000]^T$	-2.0000	$[-1.0000, 1.0000, -1.0000]^T$
3	$[-3.0000, 4.0000, -3.0000]^T$	4.0000	$[-0.7500, 1.0000, -0.7500]^T$
4	$[-2.5000, 3.5000, -2.5000]^T$	3.5000	$[-0.7143, 1.0000, -0.7143]^T$
5	$[-2.4286, 3.4286, -2.4286]^T$	3.4286	$[-0.7083, 1.0000, -0.7083]^T$
6	$[-2.4167, 3.4167, -2.4167]^T$	3.4167	$[-0.7073, 1.0000, -0.7073]^T$
7	$[-2.4146, 3.4146, -2.4146]^T$	3.4146	$[-0.7071, 1.0000, -0.7071]^T$

Conclusion: Characteristic equation is $(2 - \lambda)[(2 - \lambda)^2 - 2] = 0$. The exact eigenvalues are $\lambda = 2, 2 \pm \sqrt{2}$. The method converges to the dominant eigenvalue:

$$\lambda_1 = 2 + \sqrt{2} \approx 3.41421, \quad \mathbf{X}_1 = \left[\frac{1}{\sqrt{2}}, -1, \frac{1}{\sqrt{2}} \right]^T \approx [0.7071, -1.0000, 0.7071]^T$$

Incomplete Lecture Exercise 2 (Page 15 of Notes)

Find the largest eigenvalue and corresponding eigenvector of:

$$A = \begin{bmatrix} 1 & 3 & -1 \\ 3 & 2 & 4 \\ -1 & 4 & 10 \end{bmatrix}$$

Complete Rigorous Solution

We take initial guess $X_0 = [1, 1, 1]^T$.

k	$Y_k = AX_{k-1}$	$m_k = \max Y_k $	Normalized $X_k = Y_k/m_k$
1	$[3.0000, 9.0000, 13.0000]^T$	13.0000	$[0.2308, 0.6923, 1.0000]^T$
2	$[1.3077, 6.0769, 12.5385]^T$	12.5385	$[0.1043, 0.4847, 1.0000]^T$
3	$[0.5583, 5.2822, 11.8344]^T$	11.8344	$[0.0472, 0.4463, 1.0000]^T$
4	$[0.3862, 5.0343, 11.7382]^T$	11.7382	$[0.0329, 0.4289, 1.0000]^T$
5	$[0.3197, 4.9566, 11.6826]^T$	11.6826	$[0.0274, 0.4243, 1.0000]^T$
6	$[0.3002, 4.9310, 11.6697]^T$	11.6697	$[0.0257, 0.4225, 1.0000]^T$
7	$[0.2929, 4.9223, 11.6643]^T$	11.6643	$[0.0251, 0.4220, 1.0000]^T$

Conclusion: The dominant eigenvalue correct to 3 decimal places is $\lambda_1 \approx \mathbf{11.662}$, with eigenvector $\mathbf{X}_1 \approx [0.0251, 0.4220, 1.0000]^T$. (Exact: $\lambda_1 \approx 11.66199$).

Incomplete Lecture Exercise 3 (Page 15 of Notes)

Find the largest eigenvalue and corresponding eigenvector of:

$$A = \begin{bmatrix} 1 & 6 & 1 \\ 4 & 2 & 0 \\ 0 & 0 & 3 \end{bmatrix}$$

Complete Rigorous Solution

Notice the upper block-triangular structure:

$$\det(A - \lambda I) = (3 - \lambda) \det \begin{bmatrix} 1 - \lambda & 6 \\ 4 & 2 - \lambda \end{bmatrix} = (3 - \lambda)[(1 - \lambda)(2 - \lambda) - 24] = (3 - \lambda)(\lambda^2 - 3\lambda - 22) = 0$$

The roots are $\lambda = 3$ and $\lambda = \frac{3 \pm \sqrt{9+88}}{2} = \frac{3 \pm \sqrt{97}}{2} \approx \frac{3 \pm 9.848858}{2}$. Thus:

$$\lambda_1 = \frac{3 + \sqrt{97}}{2} \approx \mathbf{6.424429}, \quad \lambda_2 = \frac{3 - \sqrt{97}}{2} \approx -3.424429, \quad \lambda_3 = 3$$

The dominant eigenvalue is $\lambda_1 \approx \mathbf{6.424429}$. Applying the Power Method starting with $X_0 = [1, 1, 1]^T$:

k	$Y_k = AX_{k-1}$	$m_k = \max Y_k $	Normalized $X_k = Y_k/m_k$
1	$[8.0000, 6.0000, 3.0000]^T$	8.0000	$[1.0000, 0.7500, 0.3750]^T$
2	$[5.8750, 5.5000, 1.1250]^T$	5.8750	$[1.0000, 0.9362, 0.1915]^T$
3	$[6.8085, 5.8723, 0.5745]^T$	6.8085	$[1.0000, 0.8625, 0.0844]^T$
4	$[6.2594, 5.7250, 0.2531]^T$	6.2594	$[1.0000, 0.9146, 0.0404]^T$
5	$[6.5282, 5.8293, 0.1213]^T$	6.5282	$[1.0000, 0.8929, 0.0186]^T$
6	$[6.3762, 5.7859, 0.0558]^T$	6.3762	$[1.0000, 0.9074, 0.0087]^T$
7	$[6.4533, 5.8148, 0.0262]^T$	6.4533	$[1.0000, 0.9011, 0.0041]^T$

As $k \rightarrow \infty$, $m_k \rightarrow \mathbf{6.4244}$ and $X_k \rightarrow [1.0000, 0.9041, 0.0000]^T$.

Incomplete Lecture Exercise 4 (Page 15 of Notes)

Find the largest eigenvalue and corresponding eigenvector of:

$$A = \begin{bmatrix} 3 & 2 & 1 \\ 2 & 3 & 2 \\ 1 & 2 & 3 \end{bmatrix}$$

Complete Rigorous Solution

Matrix A is symmetric. The characteristic polynomial is:

$$\det(A - \lambda I) = -(\lambda - 2)(\lambda^2 - 7\lambda + 4) = 0$$

The exact eigenvalues are $\lambda = 2$ and $\lambda = \frac{7 \pm \sqrt{49-16}}{2} = \frac{7 \pm \sqrt{33}}{2}$. The dominant eigenvalue is:

$$\lambda_1 = \frac{7 + \sqrt{33}}{2} \approx \mathbf{6.372281}$$

Iterating with $X_0 = [1, 1, 1]^T$:

k	$Y_k = AX_{k-1}$	$m_k = \max Y_k $	Normalized $X_k = Y_k/m_k$
1	$[6.0000, 7.0000, 6.0000]^T$	7.0000	$[0.8571, 1.0000, 0.8571]^T$
2	$[5.4286, 6.4286, 5.4286]^T$	6.4286	$[0.8444, 1.0000, 0.8444]^T$
3	$[5.3778, 6.3778, 5.3778]^T$	6.3778	$[0.8432, 1.0000, 0.8432]^T$
4	$[5.3728, 6.3728, 5.3728]^T$	6.3728	$[0.8431, 1.0000, 0.8431]^T$
5	$[5.3723, 6.3723, 5.3723]^T$	6.3723	$[0.8431, 1.0000, 0.8431]^T$

Conclusion: Converges to $\lambda_1 = \frac{7+\sqrt{33}}{2} \approx \mathbf{6.37228}$ and eigenvector $\mathbf{X}_1 = \left[\frac{\sqrt{33}-3}{4}, 1, \frac{\sqrt{33}-3}{4} \right]^T \approx [0.8431, 1.0000, 0.8431]^T$.

4 Calculus of Finite Differences & Operator Algebra

4.1 Fundamental Operators

Let $y = f(x)$ be tabulated at equispaced nodes $x_i = x_0 + ih$ ($i = 0, 1, \dots, n$) with step size $h > 0$. We define the following core operators:

Operator Name	Symbol & Definition	Operator Relation
Forward Difference	$\Delta y(x) = y(x+h) - y(x)$	$\Delta \equiv E - 1$
Backward Difference	$\nabla y(x) = y(x) - y(x-h)$	$\nabla \equiv 1 - E^{-1}$
Shift Operator	$Ey(x) = y(x+h)$	$E \equiv 1 + \Delta \equiv e^{hD}$
Central Difference	$\delta y(x) = y(x+h/2) - y(x-h/2)$	$\delta \equiv E^{1/2} - E^{-1/2}$
Averaging Operator	$\mu y(x) = \frac{1}{2}[y(x+h/2) + y(x-h/2)]$	$\mu \equiv \frac{1}{2}(E^{1/2} + E^{-1/2})$
Differential Operator	$Dy(x) = \frac{dy}{dx}$	$hD \equiv \ln E \equiv \ln(1 + \Delta)$

4.2 Difference Tables

4.2.1 Forward Difference Table

$$\Delta^n y_r = \Delta^{n-1} y_{r+1} - \Delta^{n-1} y_r$$

x	y	Δy	$\Delta^2 y$	$\Delta^3 y$	$\Delta^4 y$
x_0	y_0	Δy_0			
x_1	y_1	Δy_1	$\Delta^2 y_0$	$\Delta^3 y_0$	
x_2	y_2	Δy_2	$\Delta^2 y_1$	$\Delta^3 y_1$	$\Delta^4 y_0$
x_3	y_3	Δy_3	$\Delta^2 y_2$		
x_4	y_4				

4.2.2 Backward Difference Table

$$\nabla^n y_r = \nabla^{n-1} y_r - \nabla^{n-1} y_{r-1}$$

x	y	∇y	$\nabla^2 y$	$\nabla^3 y$	$\nabla^4 y$
x_0	y_0				
x_1	y_1	∇y_1			
x_2	y_2	∇y_2	$\nabla^2 y_2$		
x_3	y_3	∇y_3	$\nabla^2 y_3$	$\nabla^3 y_3$	
x_4	y_4	∇y_4	$\nabla^2 y_4$	$\nabla^3 y_4$	$\nabla^4 y_4$

4.3 Rigorous Proofs of Operator Identities

Operator Identity 1: $\mu\delta = \sinh(hD)$

Prove that $\mu\delta \equiv \sinh(hD)$.

Proof. By definition, $\mu = \frac{1}{2}(E^{1/2} + E^{-1/2})$ and $\delta = E^{1/2} - E^{-1/2}$.

$$\mu\delta = \frac{1}{2} \left(E^{1/2} + E^{-1/2} \right) \left(E^{1/2} - E^{-1/2} \right) = \frac{1}{2} (E - E^{-1})$$

Using $E = e^{hD}$ and $E^{-1} = e^{-hD}$:

$$\mu\delta = \frac{1}{2} (e^{hD} - e^{-hD}) = \sinh(hD)$$

□

Operator Identity 2: $\Delta + \nabla \equiv \frac{\Delta}{\nabla} - \frac{\nabla}{\Delta}$

Prove that $\Delta + \nabla \equiv \frac{\Delta}{\nabla} - \frac{\nabla}{\Delta}$.

Proof. Expressing in terms of shift operator E : $\Delta = E - 1$ and $\nabla = 1 - E^{-1} = \frac{E-1}{E}$.

$$\frac{\Delta}{\nabla} = \frac{E-1}{(E-1)/E} = E, \quad \frac{\nabla}{\Delta} = \frac{(E-1)/E}{E-1} = \frac{1}{E} = E^{-1}$$

Therefore:

$$\text{RHS} = \frac{\Delta}{\nabla} - \frac{\nabla}{\Delta} = E - E^{-1} = (E-1) + (1-E^{-1}) = \Delta + \nabla = \text{LHS}$$

□

Operator Identity 3: $\Delta = \frac{1}{2}\delta^2 + \delta\sqrt{1 + \frac{1}{4}\delta^2}$ (Incomplete on Page 19)

Prove that $\Delta \equiv \frac{1}{2}\delta^2 + \delta\sqrt{1 + \frac{1}{4}\delta^2}$.

Proof. Notice that:

$$1 + \frac{1}{4}\delta^2 = 1 + \frac{1}{4}(E^{1/2} - E^{-1/2})^2 = 1 + \frac{1}{4}(E - 2 + E^{-1}) = \frac{E + 2 + E^{-1}}{4} = \left(\frac{E^{1/2} + E^{-1/2}}{2} \right)^2 = \mu^2$$

Taking the square root yields $\sqrt{1 + \frac{1}{4}\delta^2} = \mu = \frac{1}{2}(E^{1/2} + E^{-1/2})$. Now evaluate the right-hand side:

$$\begin{aligned} \text{RHS} &= \frac{1}{2}\delta^2 + \delta\mu = \frac{1}{2}(E - 2 + E^{-1}) + \frac{1}{2}(E^{1/2} - E^{-1/2})(E^{1/2} + E^{-1/2}) \\ &= \frac{1}{2}(E - 2 + E^{-1}) + \frac{1}{2}(E - E^{-1}) \\ &= \frac{1}{2}(2E - 2) = E - 1 = \Delta = \text{LHS} \end{aligned}$$

□

Operator Identity 4: $\Delta^n(1/x)$ (Incomplete on Page 19)

Evaluate $\Delta^2\left(\frac{1}{x}\right)$ and state the general formula for $\Delta^n\left(\frac{1}{x}\right)$.

Proof. For step length h :

$$\Delta\left(\frac{1}{x}\right) = \frac{1}{x+h} - \frac{1}{x} = \frac{-h}{x(x+h)}$$

$$\Delta^2\left(\frac{1}{x}\right) = \Delta\left[\frac{-h}{x(x+h)}\right] = \frac{-h}{(x+h)(x+2h)} - \left[\frac{-h}{x(x+h)}\right] = -h\left[\frac{x - (x+2h)}{x(x+h)(x+2h)}\right] = \frac{2h^2}{x(x+h)(x+2h)}$$

By induction, the general n -th difference is:

$$\Delta^n\left(\frac{1}{x}\right) = \frac{(-1)^n n! h^n}{x(x+h)(x+2h)\dots(x+nh)} \quad (30)$$

For unit step length $h = 1$: $\Delta^2\left(\frac{1}{x}\right) = \frac{2}{x(x+1)(x+2)}$. □

Operator Identity 5: Exponential Series Sum (Incomplete on Page 27)

Prove that:

$$e^x \left\{ y_0 + x\Delta y_0 + \frac{x^2}{2!}\Delta^2 y_0 + \dots \right\} = y_0 + xy_1 + \frac{x^2}{2!}y_2 + \dots \quad (31)$$

Proof. Express the inner series using operator notation:

$$\sum_{n=0}^{\infty} \frac{x^n \Delta^n}{n!} y_0 = e^{x\Delta} y_0$$

Multiplying by e^x :

$$\text{LHS} = e^x \cdot e^{x\Delta} y_0 = e^{x(1+\Delta)} y_0$$

Since $1 + \Delta = E$:

$$\text{LHS} = e^{xE} y_0 = \sum_{n=0}^{\infty} \frac{(xE)^n}{n!} y_0 = \sum_{n=0}^{\infty} \frac{x^n}{n!} E^n y_0 = \sum_{n=0}^{\infty} \frac{x^n}{n!} y_n = y_0 + xy_1 + \frac{x^2}{2!}y_2 + \dots = \text{RHS}$$

□

Operator Identity 6: Alternating Difference Identity (Incomplete on Page 28)

Prove that:

$$y_x - \Delta^2 y_x + \Delta^3 y_x - \Delta^5 y_x + \Delta^6 y_x - \Delta^8 y_x + \dots = y_x - \Delta^2 y_{x-1} + \Delta^4 y_{x-2} - \Delta^6 y_{x-3} + \dots \quad (32)$$

Proof. Consider the left-hand side operator acting on y_x :

$$L = 1 - \Delta^2 + \Delta^3 - \Delta^5 + \Delta^6 - \Delta^8 + \dots = (1 - \Delta^2) + \Delta^3(1 - \Delta^2) + \Delta^6(1 - \Delta^2) + \dots$$

Factoring $(1 - \Delta^2)$:

$$L = (1 - \Delta^2)(1 + \Delta^3 + \Delta^6 + \dots) = \frac{1 - \Delta^2}{1 - \Delta^3}$$

Factoring numerator and denominator:

$$L = \frac{(1 - \Delta)(1 + \Delta)}{(1 - \Delta)(1 + \Delta + \Delta^2)} = \frac{1 + \Delta}{1 + \Delta + \Delta^2} = \frac{E}{E + \Delta^2}$$

Dividing numerator and denominator by E :

$$L = \frac{1}{1 + \Delta^2 E^{-1}} = 1 - \Delta^2 E^{-1} + (\Delta^2 E^{-1})^2 - (\Delta^2 E^{-1})^3 + \dots = 1 - \Delta^2 E^{-1} + \Delta^4 E^{-2} - \Delta^6 E^{-3} + \dots$$

Operating on y_x :

$$Ly_x = y_x - \Delta^2 y_{x-1} + \Delta^4 y_{x-2} - \Delta^6 y_{x-3} + \dots = \text{RHS}$$

□

Operator Identity 7: Midpoint Curvature Identity (Incomplete on Page 32)

If $\Delta^3 y_x = 0$, prove that:

$$y_{x+1/2} = \frac{1}{2}(y_x + y_{x+1}) - \frac{1}{16}(\Delta^2 y_x + \Delta^2 y_{x+1}) \quad (33)$$

Proof. Recall $y_{x+1/2} = E^{1/2} y_x$. Using $E = 1 + \Delta$:

$$E^{1/2} = (1 + \Delta)^{1/2} = 1 + \frac{1}{2}\Delta - \frac{1}{8}\Delta^2 + \frac{1}{16}\Delta^3 - \dots$$

Since $\Delta^3 y_x = 0$, all terms of order Δ^3 and higher vanish:

$$\text{LHS} = y_{x+1/2} = \left(1 + \frac{1}{2}\Delta - \frac{1}{8}\Delta^2\right) y_x$$

Now express the RHS in terms of Δ :

$$\frac{1}{2}(y_x + y_{x+1}) = \frac{1}{2}(1 + E)y_x = \frac{1}{2}(2 + \Delta)y_x = \left(1 + \frac{1}{2}\Delta\right) y_x$$

$$\frac{1}{16}(\Delta^2 y_x + \Delta^2 y_{x+1}) = \frac{1}{16}\Delta^2(1 + E)y_x = \frac{1}{16}\Delta^2(2 + \Delta)y_x = \frac{1}{8}\Delta^2 y_x + \frac{1}{16}\Delta^3 y_x = \frac{1}{8}\Delta^2 y_x$$

Subtracting:

$$\text{RHS} = \left(1 + \frac{1}{2}\Delta\right) y_x - \frac{1}{8}\Delta^2 y_x = \left(1 + \frac{1}{2}\Delta - \frac{1}{8}\Delta^2\right) y_x = \text{LHS}$$

□

Operator Identity 8: Binomial Shift Identity (Page 26 of Notes)

Prove that:

$$\begin{aligned} y_x - \frac{1}{8}\Delta^2 y_{x-1} + \frac{1 \cdot 3}{8 \cdot 16}\Delta^4 y_{x-2} - \frac{1 \cdot 3 \cdot 5}{8 \cdot 16 \cdot 24}\Delta^6 y_{x-3} + \dots \\ = y_{x+1/2} - \frac{1}{2}\Delta y_{x+1/2} + \frac{1}{4}\Delta^2 y_{x+1/2} - \frac{1}{8}\Delta^3 y_{x+1/2} + \dots \end{aligned} \quad (34)$$

Proof. Express the left-hand side using the backward shift operator E^{-1} :

$$\begin{aligned} \text{LHS} &= \left[1 - \frac{1}{8}\Delta^2 E^{-1} + \frac{1 \cdot 3}{8 \cdot 16}\Delta^4 E^{-2} - \frac{1 \cdot 3 \cdot 5}{8 \cdot 16 \cdot 24}\Delta^6 E^{-3} + \dots\right] y_x \\ &= \left[1 - \frac{1}{2}\left(\frac{\Delta^2 E^{-1}}{4}\right) + \frac{\frac{1}{2} \cdot \frac{3}{2}}{2!}\left(\frac{\Delta^2 E^{-1}}{4}\right)^2 - \frac{\frac{1}{2} \cdot \frac{3}{2} \cdot \frac{5}{2}}{3!}\left(\frac{\Delta^2 E^{-1}}{4}\right)^3 + \dots\right] y_x \end{aligned}$$

This is precisely the binomial expansion of $(1 + u)^{-1/2}$ for $u = \frac{\Delta^2 E^{-1}}{4}$:

$$\begin{aligned} \text{LHS} &= \left(1 + \frac{\Delta^2}{4E}\right)^{-1/2} y_x = \left(\frac{4E + \Delta^2}{4E}\right)^{-1/2} y_x \\ &= \frac{(4E + \Delta^2)^{-1/2}}{(4E)^{-1/2}} y_x = 2E^{1/2}(4E + \Delta^2)^{-1/2} y_x = 2(4E + \Delta^2)^{-1/2} y_{x+1/2} \end{aligned}$$

Using the operator equivalence $E \equiv 1 + \Delta$:

$$4E + \Delta^2 = 4(1 + \Delta) + \Delta^2 = \Delta^2 + 4\Delta + 4 = (\Delta + 2)^2$$

Substituting into the expression:

$$\text{LHS} = 2 [(\Delta + 2)^2]^{-1/2} y_{x+1/2} = 2(\Delta + 2)^{-1} y_{x+1/2} = \frac{2}{2(1 + \frac{\Delta}{2})} y_{x+1/2} = \left(1 + \frac{\Delta}{2}\right)^{-1} y_{x+1/2}$$

Expanding $(1 + \Delta/2)^{-1}$ as a geometric series:

$$\begin{aligned} \text{LHS} &= \left[1 - \frac{\Delta}{2} + \left(\frac{\Delta}{2}\right)^2 - \left(\frac{\Delta}{2}\right)^3 + \dots\right] y_{x+1/2} \\ &= y_{x+1/2} - \frac{1}{2}\Delta y_{x+1/2} + \frac{1}{4}\Delta^2 y_{x+1/2} - \frac{1}{8}\Delta^3 y_{x+1/2} + \dots = \text{RHS} \end{aligned}$$

□

Operator Identity 9: Alternating Difference Series Summation (Page 29 of Notes)

Evaluate the sum of the operator series:

$$S = 1 \cdot 2\Delta x^n - 2 \cdot 3\Delta^2 x^n + 3 \cdot 4\Delta^3 x^n - 4 \cdot 5\Delta^4 x^n + \dots \quad (35)$$

Proof. Operating by Δ on S :

$$\begin{aligned} S &= 1 \cdot 2\Delta x^n - 2 \cdot 3\Delta^2 x^n + 3 \cdot 4\Delta^3 x^n - 4 \cdot 5\Delta^4 x^n + \dots \\ \Delta S &= \quad \quad \quad + 1 \cdot 2\Delta^2 x^n - 2 \cdot 3\Delta^3 x^n + 3 \cdot 4\Delta^4 x^n - \dots \end{aligned}$$

Adding the two equations:

$$\begin{aligned} (1 + \Delta)S &= 1 \cdot 2\Delta x^n - (6 - 2)\Delta^2 x^n + (12 - 6)\Delta^3 x^n - (20 - 12)\Delta^4 x^n + \dots \\ &= 2\Delta x^n - 4\Delta^2 x^n + 6\Delta^3 x^n - 8\Delta^4 x^n + \dots \\ &= 2\Delta [1 - 2\Delta + 3\Delta^2 - 4\Delta^3 + \dots] x^n \end{aligned}$$

Recall that $(1 + \Delta)^{-2} = 1 - 2\Delta + 3\Delta^2 - 4\Delta^3 + \dots$. Therefore:

$$(1 + \Delta)S = 2\Delta(1 + \Delta)^{-2} x^n \implies S = 2\Delta(1 + \Delta)^{-3} x^n$$

Using $E = 1 + \Delta$ and $\Delta = E - 1$:

$$S = 2(E - 1)E^{-3} x^n = 2(E^{-2} - E^{-3}) x^n = 2[E^{-2} x^n - E^{-3} x^n]$$

For step size h , $E^{-k} f(x) = f(x - kh)$:

$$S = 2[(x - 2h)^n - (x - 3h)^n] \quad (36)$$

For unit step length $h = 1$: $S = 2[(x - 2)^n - (x - 3)^n]$. □

4.4 Fundamental Theorem of Finite Differences

Fundamental Theorem of Finite Differences

The n -th forward difference of a polynomial $P_n(x)$ of degree n with step size h is constant and equal to $a_0 n! h^n$, and all higher differences are identically zero:

$$\Delta^n P_n(x) = a_0 n! h^n = \text{constant}, \quad \Delta^{n+k} P_n(x) = 0 \quad (k \geq 1) \quad (37)$$

4.5 Missing Value Problems

Classroom Problem: Single Missing Value with Error Analysis (Page 22 of Notes)

Find the missing entry y_3 in the following data table:

x	0	1	2	3	4
y	1	3	9	y_3	81

Explain why the interpolated value differs from $3^3 = 27$.

Step-by-Step Solution

Since four values are given ($y_0 = 1, y_1 = 3, y_2 = 9, y_4 = 81$), the underlying function can be modeled by a polynomial of degree 3. By the Fundamental Theorem of Finite Differences, the fourth differences vanish:

$$\Delta^4 y_0 = 0 \iff (E - 1)^4 y_0 = 0$$

Expanding the binomial operator $(E - 1)^4 = E^4 - 4E^3 + 6E^2 - 4E + 1$:

$$y_4 - 4y_3 + 6y_2 - 4y_1 + y_0 = 0$$

Isolating the missing entry y_3 :

$$y_3 = \frac{1}{4} (y_4 + 6y_2 - 4y_1 + y_0)$$

Substitute the tabulated values:

$$y_3 = \frac{1}{4} [81 + 6(9) - 4(3) + 1] = \frac{1}{4} [81 + 54 - 12 + 1] = \frac{124}{4} = \mathbf{31}$$

Mathematical Note on Discrepancy (Recorded in Lecture Notes): Notice that the tabulated values $y = [1, 3, 9, \dots, 81]$ represent the exponential function $f(x) = 3^x$, for which $f(3) = 3^3 = 27$. However, the finite difference method assumes $f(x)$ is a cubic polynomial. Because 3^x is *not* a polynomial, the cubic polynomial passing through $(0, 1), (1, 3), (2, 9), (4, 81)$ produces $y_3 = 31$, incurring a truncation discrepancy of $31 - 27 = 4$.

Incomplete Lecture Problem: Two Missing Values (Pages 22–23 of Notes)

Find the missing entries y_1 and y_3 in the following data table:

x	0	1	2	3	4	5
y	1	y_1	8	y_3	19	31

Complete Rigorous Solution

Since four functional values are given ($y_0 = 1, y_2 = 8, y_4 = 19, y_5 = 31$), the underlying function can be represented by a polynomial of degree 3. By the Fundamental Theorem of Finite Differences, the fourth differences are zero:

$$\Delta^4 y_0 = 0 \iff (E - 1)^4 y_0 = 0 \quad (38)$$

$$\Delta^4 y_1 = 0 \iff (E - 1)^4 y_1 = 0 \quad (39)$$

Expanding the binomial operator $(E - 1)^4 = E^4 - 4E^3 + 6E^2 - 4E + 1$:

$$y_4 - 4y_3 + 6y_2 - 4y_1 + y_0 = 0 \quad (40)$$

$$y_5 - 4y_4 + 6y_3 - 4y_2 + y_1 = 0 \quad (41)$$

Substitute the known values into (40):

$$19 - 4y_3 + 6(8) - 4y_1 + 1 = 0 \implies 68 - 4y_1 - 4y_3 = 0 \implies \mathbf{y_1 + y_3 = 17}$$

Substitute into (41):

$$31 - 4(19) + 6y_3 - 4(8) + y_1 = 0 \implies 31 - 76 + 6y_3 - 32 + y_1 = 0 \implies \mathbf{y_1 + 6y_3 = 77}$$

Subtracting the two equations:

$$(y_1 + 6y_3) - (y_1 + y_3) = 77 - 17 \implies 5y_3 = 60 \implies \mathbf{y_3 = 12}$$

Substituting back:

$$y_1 = 17 - 12 \implies \mathbf{y_1 = 5}$$

Verification via Difference Table: The complete dataset is $y = [1, 5, 8, 12, 19, 31]$.

- First differences Δy : 4, 3, 4, 7, 12.
- Second differences $\Delta^2 y$: -1, 1, 3, 5.
- Third differences $\Delta^3 y$: 2, 2, 2 (constant!).
- Fourth differences $\Delta^4 y$: 0, 0 (verified!).

Final Answer: $\mathbf{y_1 = 5, \quad y_3 = 12}$.

4.6 Factorial Notation & Polynomial Conversion

Factorial Polynomial

The factorial polynomial of degree n with step length $h = 1$ is defined as:

$$[x]^n = x(x-1)(x-2)\dots(x-n+1), \quad [x]^0 = 1 \quad (42)$$

Its forward difference mirrors standard differential calculus:

$$\Delta[x]^n = n[x]^{n-1}, \quad \Delta^{-1}[x]^n = \frac{[x]^{n+1}}{n+1} + C \quad (43)$$

Classroom Problem 1: Interval of Differencing (Page 24 of Notes)

Find the interval of differencing h if $\Delta x^2 = x + \frac{1}{4}$.

Step-by-Step Solution

By definition of the forward difference operator with step length h :

$$\Delta x^2 = (x+h)^2 - x^2 = x^2 + 2hx + h^2 - x^2 = 2hx + h^2$$

We are given that $\Delta x^2 = x + \frac{1}{4}$. Equating coefficients of like powers of x :

$$\begin{aligned} x^1 : \quad 2h &= 1 \implies h = \frac{1}{2} \\ x^0 : \quad h^2 &= \frac{1}{4} \implies h = \pm \frac{1}{2} \end{aligned}$$

Both conditions are consistently satisfied by:

$$\mathbf{h} = \frac{1}{2}$$

Hence, the required interval of differencing is $\mathbf{h} = 1/2$.

Classroom Problem 2: Factorial Form of Cubic Polynomial (Page 24 of Notes)

Express $y = 2x^3 - 3x^2 + 3x - 1$ in factorial notation and determine its successive differences $\Delta y, \Delta^2 y, \Delta^3 y, \Delta^4 y$.

Solution via Synthetic Division

Let $y = A[x]^3 + B[x]^2 + C[x] + D$. At $x = 0$, $D = -1$. We find coefficients C, B, A by dividing successively by $x = 1, 2$:

1	2	-3	3	-1
		2	-1	2
	2	-1	2 = C	-1 = D
2		4		
	2 = A	3 = B		

Hence, the factorial representation is:

$$\boxed{y = 2[x]^3 + 3[x]^2 + 2[x] - 1} \quad (44)$$

Successive Differences (for $h = 1$):

$$\Delta y = 2 \cdot 3[x]^2 + 3 \cdot 2[x] + 2 = 6[x]^2 + 6[x] + 2$$

$$\Delta^2 y = 6 \cdot 2[x] + 6 = 12[x] + 6$$

$$\Delta^3 y = 12 \cdot 1 = 12$$

$$\Delta^4 y = 0$$

Incomplete Lecture Exercise (Page 25 of Notes)

Express $f(x) = 11x^4 + 5x^3 + x - 15$ in factorial notation and compute all successive differences.

Complete Rigorous Solution

Notice the polynomial includes a missing second-degree term:

$$f(x) = 11x^4 + 5x^3 + 0x^2 + 1x - 15$$

Let $f(x) = A[x]^4 + B[x]^3 + C[x]^2 + D[x] + E$. Setting $x = 0$ immediately gives $\mathbf{E} = -\mathbf{15}$. We determine coefficients D, C, B, A using sequential Horner synthetic division:

	11	5	0	1
1		11	16	16
2	11	16	16	17 = D
		22	76	
3	11	38	92 = C	
		33		
	11 = A	71 = B		

Thus, the factorial representation is:

$$f(x) = 11[x]^4 + 71[x]^3 + 92[x]^2 + 17[x] - 15 \quad (45)$$

Successive Differences:

$$\begin{aligned} \Delta f(x) &= 44[x]^3 + 213[x]^2 + 184[x] + 17 \\ \Delta^2 f(x) &= 132[x]^2 + 426[x] + 184 \\ \Delta^3 f(x) &= 264[x] + 426 \\ \Delta^4 f(x) &= 264 \\ \Delta^5 f(x) &= 0 \end{aligned}$$

4.7 Summation of Series using Finite Differences

Incomplete Lecture Problem (Page 29 of Notes)

Find the sum of n terms of the series:

$$S_n = 2 \cdot 5 + 5 \cdot 8 + 8 \cdot 11 + 11 \cdot 14 + \dots$$

Complete Rigorous Solution

The general r -th term is:

$$T_r = (3r - 1)(3r + 2) = 9r^2 + 3r - 2$$

Converting r^2 into factorial notation ($r^2 = [r]^2 + [r]$):

$$T_r = 9([r]^2 + [r]) + 3[r] - 2 = 9[r]^2 + 12[r] - 2$$

Using the antidifference summation formula $\sum_{r=1}^n T_r = [\Delta^{-1}T_r]_1^{n+1}$:

$$\begin{aligned} S_n &= \left[9\frac{[r]^3}{3} + 12\frac{[r]^2}{2} - 2[r] \right]_1^{n+1} = [3[r]^3 + 6[r]^2 - 2[r]]_1^{n+1} \\ &= \{3(n+1)n(n-1) + 6(n+1)n - 2(n+1)\} - \{0 + 0 - 2(1)\} \\ &= (n+1)[3n^2 - 3n + 6n - 2] + 2 = (n+1)(3n^2 + 3n - 2) + 2 \\ &= 3n^3 + 6n^2 + n \end{aligned}$$

$$\boxed{S_n = n(3n^2 + 6n + 1)} \quad (46)$$

Verification: For $n = 1$: $S_1 = 1(3 + 6 + 1) = 10 = 2 \cdot 5$. For $n = 2$: $S_2 = 2(12 + 12 + 1) = 50 = 10 + 40$. For $n = 3$: $S_3 = 3(27 + 18 + 1) = 138 = 50 + 88$. Matches perfectly!

Incomplete Lecture Problem (Page 30 of Notes)

Find the n -th term of the sequence: 1, 4, 10, 20, 35, 56, ...

Complete Rigorous Solution

Construct the difference table for the sequence $u_1, u_2, u_3, \dots$:

- u : 1, 4, 10, 20, 35, 56
- Δu : 3, 6, 10, 15, 21
- $\Delta^2 u$: 3, 4, 5, 6
- $\Delta^3 u$: 1, 1, 1 (constant $\Delta^3 u_1 = 1$)
- $\Delta^4 u$: 0.

Using $T_n = E^{n-1}u_1 = (1 + \Delta)^{n-1}u_1$:

$$\begin{aligned} T_n &= u_1 + (n-1)\Delta u_1 + \frac{(n-1)(n-2)}{2!}\Delta^2 u_1 + \frac{(n-1)(n-2)(n-3)}{3!}\Delta^3 u_1 \\ &= 1 + 3(n-1) + \frac{3}{2}(n-1)(n-2) + \frac{1}{6}(n-1)(n-2)(n-3) \end{aligned}$$

Combining terms over common denominator 6:

$$\begin{aligned} T_n &= \frac{1}{6} [6 + 18(n-1) + 9(n^2 - 3n + 2) + (n^3 - 6n^2 + 11n - 6)] \\ &= \frac{1}{6} [n^3 + 3n^2 + 2n] = \frac{n(n+1)(n+2)}{6} = \binom{n+2}{3} \end{aligned}$$

$$\boxed{T_n = \frac{n(n+1)(n+2)}{6} = \binom{n+2}{3}} \quad (47)$$

Incomplete Lecture Problem: Sum of Squares of First n Natural Numbers (Page 30 of Notes)

Evaluate the sum of the series using finite difference methods:

$$S_n = 1^2 + 2^2 + 3^2 + \dots + n^2 \quad (48)$$

Solution via Factorial Polynomials

Convert r^2 into factorial polynomials:

$$r^2 = [r]^2 + [r]$$

Applying the summation formula $\sum_{r=1}^n T_r = [\Delta^{-1}T_r]_1^{n+1}$:

$$\begin{aligned} S_n &= \sum_{r=1}^n ([r]^2 + [r]) = \left[\frac{[r]^3}{3} + \frac{[r]^2}{2} \right]_1^{n+1} \\ &= \left\{ \frac{(n+1)n(n-1)}{3} + \frac{(n+1)n}{2} \right\} - \{0 + 0\} \\ &= \frac{n(n+1)}{6} [2(n-1) + 3] = \frac{n(n+1)(2n+1)}{6} \end{aligned}$$

$$\boxed{\sum_{r=1}^n r^2 = \frac{n(n+1)(2n+1)}{6}} \quad (49)$$

Classroom Problem: Sum of Cubes via Operator Formula (Page 31 of Notes)

Evaluate the sum of the first n cubes:

$$S_n = 1^3 + 2^3 + 3^3 + \cdots + n^3 \quad (50)$$

using the finite difference summation operator $\frac{(1+\Delta)^n - 1}{\Delta} y_0$.

Complete Operator Derivation Recorded on Page 31

Let $y_r = (r+1)^3$ for $r = 0, 1, \dots, n-1$, so that $S_n = y_0 + y_1 + \dots + y_{n-1}$. Expressing in terms of shift operator E :

$$S_n = (1 + E + E^2 + \dots + E^{n-1})y_0 = \frac{E^n - 1}{E - 1}y_0$$

Since $E = 1 + \Delta$ and $E - 1 = \Delta$:

$$S_n = \frac{(1 + \Delta)^n - 1}{\Delta}y_0$$

Expanding $(1 + \Delta)^n$ using the binomial theorem:

$$\begin{aligned} S_n &= \frac{1}{\Delta} \left[1 + n\Delta + \frac{n(n-1)}{2!}\Delta^2 + \frac{n(n-1)(n-2)}{3!}\Delta^3 + \frac{n(n-1)(n-2)(n-3)}{4!}\Delta^4 + \dots - 1 \right] y_0 \\ &= ny_0 + \frac{n(n-1)}{2!}\Delta y_0 + \frac{n(n-1)(n-2)}{3!}\Delta^2 y_0 + \frac{n(n-1)(n-2)(n-3)}{4!}\Delta^3 y_0 + \dots \end{aligned}$$

Difference table for $y_x = (x+1)^3$:

x	$y = (x+1)^3$	Δy	$\Delta^2 y$	$\Delta^3 y$	$\Delta^4 y$
0	1 = y_0				
1	8	7 = Δy_0	12 = $\Delta^2 y_0$		
2	27	19	18	6 = $\Delta^3 y_0$	0 = $\Delta^4 y_0$
3	64	37	24	6	0

Substituting $y_0 = 1$, $\Delta y_0 = 7$, $\Delta^2 y_0 = 12$, $\Delta^3 y_0 = 6$, and $\Delta^k y_0 = 0$ ($k \geq 4$):

$$\begin{aligned} S_n &= n(1) + \frac{n(n-1)}{2}(7) + \frac{n(n-1)(n-2)}{6}(12) + \frac{n(n-1)(n-2)(n-3)}{24}(6) \\ &= n + \frac{7}{2}n(n-1) + 2n(n-1)(n-2) + \frac{1}{4}n(n-1)(n-2)(n-3) \\ &= \frac{n}{4} [4 + 14(n-1) + 8(n-1)(n-2) + (n-1)(n-2)(n-3)] \\ &= \frac{n}{4} [4 + 14n - 14 + 8(n^2 - 3n + 2) + (n^3 - 6n^2 + 11n - 6)] \\ &= \frac{n}{4} [n^3 + 2n^2 + n] = \frac{n^2(n+1)^2}{4} = \left[\frac{n(n+1)}{2} \right]^2 \end{aligned}$$

$$\boxed{\sum_{r=1}^n r^3 = \left[\frac{n(n+1)}{2} \right]^2} \quad (51)$$

5 Interpolation with Equidistant Nodes

5.1 Newton's Forward Interpolation Formula

Let $y = f(x)$ be known at $n + 1$ equidistant points $x_0, x_1, \dots, x_n$ with spacing h . Let $x = x_0 + ph$ be an interpolation point near the beginning of the table, where $p = \frac{x - x_0}{h} \in \mathbb{R}$.

Using the shift operator $E = 1 + \Delta$:

$$y(x) = E^p y_0 = (1 + \Delta)^p y_0 \quad (52)$$

Expanding $(1 + \Delta)^p$ via the binomial theorem:

$$y(x) = y_0 + p\Delta y_0 + \frac{p(p-1)}{2!}\Delta^2 y_0 + \frac{p(p-1)(p-2)}{3!}\Delta^3 y_0 + \dots + \frac{p(p-1)\dots(p-n+1)}{n!}\Delta^n y_0 \quad (53)$$

The truncation error is given by:

$$R_n(x) = \frac{p(p-1)\dots(p-n)}{(n+1)!}h^{n+1}f^{(n+1)}(\xi), \quad \xi \in (x_0, x_n) \quad (54)$$

5.2 Newton's Backward Interpolation Formula (Completed from Page 32)

When the point x lies near the end of the table (x_n) , we set:

$$x = x_n + ph \iff p = \frac{x - x_n}{h} \quad (55)$$

Using the backward difference operator $\nabla = 1 - E^{-1} \iff E^{-1} = 1 - \nabla \iff E = (1 - \nabla)^{-1}$:

$$y(x) = E^p y_n = (1 - \nabla)^{-p} y_n \quad (56)$$

Expanding $(1 - \nabla)^{-p}$ via negative binomial series:

$$(1 - \nabla)^{-p} = 1 + (-p)(-\nabla) + \frac{(-p)(-p-1)}{2!}(-\nabla)^2 + \dots = 1 + p\nabla + \frac{p(p+1)}{2!}\nabla^2 + \dots \quad (57)$$

Hence, we obtain the **Newton's Backward Interpolation Formula**:

$$y(x) = y_n + p\nabla y_n + \frac{p(p+1)}{2!}\nabla^2 y_n + \frac{p(p+1)(p+2)}{3!}\nabla^3 y_n + \dots + \frac{p(p+1)\dots(p+n-1)}{n!}\nabla^n y_n \quad (58)$$

The truncation error is:

$$R_n(x) = \frac{p(p+1)\dots(p+n)}{(n+1)!}h^{n+1}f^{(n+1)}(\xi'), \quad \xi' \in (x_0, x_n) \quad (59)$$

5.3 Selection Criteria: Forward vs. Backward Interpolation

Criterion	Newton's Forward Formula	Newton's Backward Formula
Target Point Position	Near the start of the table ($x \approx x_0$)	Near the end of the table ($x \approx x_n$)
Parameter p	$p = \frac{x - x_0}{h}$, ideally $0 \leq p < 1$	$p = \frac{x - x_n}{h}$, ideally $-1 < p \leq 0$
Operators Used	Forward differences $\Delta, \Delta^2, \dots$	Backward differences $\nabla, \nabla^2, \dots$
Reference Row	Top row of difference table (y_0)	Bottom row of difference table (y_n)
Extrapolation	Extrapolating to the left ($x < x_0$)	Extrapolating to the right ($x > x_n$)